ROSHINI KAREDLA

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Examination	Institute	CGPA/%	Year
BTech, CSE	IIT Madras	8.20	2022
Intermediate	BIEAP	98.8	2018
Matriculation	BSEAP	10	2016

Scholastic Achievements

- Secured **All India Rank 1263** in **JEE Advanced (IIT-JEE)** out of **155,000** candidates *(2018)*
- Secured **All India Rank 599** in **JEE Main** out of **1.2 million** candidates from all over India *(2018)*
- Secured **All India Rank 324** in **JEE Main** *(2018)*
- Secured State Rank **183** in AP EAMCET 2018 *(2018)*.

Experience

React Full Stack Developer *(August 2022 - July 2023)*
Manager: Vineeth Reddy *Vmock*

- Have developed a web module using frameworks **React.js** and **Laravel Lumen**
- Worked on Resume Book section backend **memory and space optimizations** reducing pdf sizes **to 5%** the original **easy accessibility** to user on frontend using different **react** libraries.
- The project included fetching data using **REST APIs**, sending emails to users, bootstrap etc.
- Worked with **MySQL** queries, **laravel collections**, manipulating data on the backend.
- Worked with React, Laravel, php, MySQL, javacript, redux tools.

Internships

Trading Analyst *(May 2021 - July 2021)*
Guide: Kanika Gupta | Summer Intern | Python, MS Office, Test Coverage *JP Morgan & Chase Co.*

- Been part of product team which delivers product to clients based on their specifications.
- Project included writing an **automated python script** for a large data index.
- **Debugged numerous scripts**, **aggregated data** and increased the efficiency by **20%**.
- Have also worked on **unit testing**, verifying and increasing **test coverage**, writing python scripts.
- Worked with python, pandas, numpy, microsoft excel and powerpoint.

BTech Project

Simulator for distributed algorithms *(Spring 2022)*
Guide: Prof. John Ebenezer Augustine | Python *IIT Madras*

- Developed a simulator which can be used to learn about distributed algorithms which include but not limited to breadth first search(**BFS**), depth first search(**DFS**), randomized leader election.
- The simulator takes care of both **synchronous and asynchronous** algorithms and also provides complex algorithms like message flooding algorithm, **crash failure**, maximal independent set(**MIS**).
- The motive of the project is to **help students** understand the working of the distributed algorithms which are not so easy to visualize.
- The source code is in python without using any external libraries which reduces computational time.
- The project stands out from remaining simulators online with respect to **availability of complex algorithms** and the option to **run customized algorithms**.

Key Course Projects

Heart Rate Monitor *(Spring 2021)*
Guide: Prof. Ayon Chakraborty | Course Assignment | Python, IoT *IIT Madras*

- Developed a heart rate montior that takes data from video of finger using torch and computes **average heart rate** as part of **IoT** course assignment.
- Utilised frequency of major changes in video after converting it to pixels for computation and leveraged videos at multiple fps for error minimization.

- Used python libraries **OpenCV2** for data collection, **pandas and numpy** to work on the data and **matplotlib** to create graphs.

Heart Rate Monitor with face recognition *(Spring 2021)*
Guide: Prof. Ayon Chakraborty | Course Project | C++, IoT *IIT Madras*

- Developed a heart rate monitor from face recognition using videos of multiple length, fps and different parts of face as an extension to heart rate monitor assignment.
- Used **C++** as main language to minimize compilation time and only **ffmpeg** as helper library to convert videos into pixelated data.
- Reduced data into groups based on repetition interval for more efficient heart rate, **upto 10%** efficient and then aggregating on entire data.

Song predictions to users on kaggle contest *(Spring 2021)*
Guide: Prof. Manikandan Narayanan | Course Project | Python *IIT Madras*

- Devised a recommendation model based on previous user ratings on **kaggle** platform.
- Used **xgboost** package to give song predictions to users based on their song ratings and components.
- Deconstructed the data to find aggregates wherever possible to make data richer for better results.

Othello bot *(Autumn 2020)*
Guide: Prof. Deepak Khemani | Course Project | C++, AI *IIT Madras*

- Created an othello bot using AI algorithms in **C++** by calculating win percentage of good moves.
- Used **min-max** and **alpha beta** algorithms to calculate the best move based on the opponent moves.
- Optimized computation by discarding moves which give lower score than previously calculated moves.

Compiler for simple C like language *(Autumn 2020)*
Guide: Prof. Rupesh Nasre | Course Project | lex, yacc, C++ *IIT Madras*

- Developed a compiler for a C type language using **Yacc** and **Lex**, that parses and generates xv6 code.
- Implemented a combination of Lexical analyzer, type checker, IR generator and semantic analyzer.

Compiler for JACK language *(Autumn 2019)*
Guide: Prof. V Kamakoti | Course Project | nand2tetris, C++ *IIT Madras*

- Developed a compiler for JACK language from scratch using **RISC V** instrctions on machine level and **C++** on higher level as a **nand2tetris** project.
- Built the compiler using boolean logic, boolean arithmetic, machine language in riscv, computer architecture for the low level compiler.
- For the high level of the compiler, built **assembler**, vm interpreter, stack arithmetic, high level language syntax analysis and code generation with the help of nand2tetris book.
- Extensively used **C++ OOPS** and **nand2tetris** package to run and test the compiler.

Technical Skills

Programming Languages: C, C++, Python, PHP, OCaml
Development: HTML, CSS, JavaScript, MySQL, React, Laravel
Software: Git, AutoCAD, LaTeX
ML Libraries: Pandas, Numpy

Extracurriculars

- Placed in **Top 10 %** in **National Mathematics Talent Competition** *(2015, 2012)*
- Achieved International **48th rank** in **International Maths Olympiad**, 2015 *(2015)*
- **District 1st** in CV Raman Talent Competition in Science *(2015, 2013, 2012).*
- Took university courses in Chinese, Sanskrit and German languages.
- Played keyboard as part of **National Cultural Appreciation** in IIT Madras.